

AI Security Automation Platform

Comprehensive Capstone Project Report

1. Executive Summary

This comprehensive report details the end-to-end design, implementation, and evaluation of the AI Security Automation Platform developed for the Week 10 Final Capstone. The platform integrates threat intelligence, machine learning anomaly detection, SOAR (Security Orchestration, Automation, and Response) orchestration, and real-time visualization into a unified, production-ready system.

2. Project Objectives

The primary objectives of this capstone project were to:

- Build a modular data ingestion layer capable of normalizing raw security logs.
- Enrich incoming log data with threat intelligence risk scores.
- Deploy a machine learning model to detect behavioral anomalies in real-time.
- Implement a SOAR engine to automatically execute defensive playbooks.
- Provide analysts with a real-time, interactive security dashboard.

3. System Architecture (Deep Dive)

The platform utilizes a highly modular, decoupled microservice architecture where each component operates independently, communicating asynchronously via file-based JSONL message queues.

3.1 Data Ingestion Layer

Implemented in `ingest_logs.py`, this module simulates streaming log ingestion from network devices. It generates normalized JSON payloads containing source IP, destination domain, byte transfers, and session data.

3.2 Threat Intelligence Enrichment

Implemented in `ti_enricher.py`, this module tails the raw logs and queries a mock Threat Intelligence API. It uses MD5 hashing to generate deterministic risk scores for IPs and domains.

3.3 Machine Learning Detection

Implemented in `ml_detector.py`, this module applies an Isolation Forest model to detect network anomalies. The model evaluates bytes sent/received, failed logins, and session duration.

3.4 SOAR Orchestration

Implemented in `soar_engine.py`, this engine acts on the final alerts. High-confidence alerts trigger automated playbooks, while medium-confidence alerts are queued for human review.

3.5 Visualization Dashboard

Implemented in dashboard.py using Streamlit, this frontend provides real-time visibility into the platform's operations.

4. Technical Stack & Implementation Details

- Language: Python 3
- Machine Learning: scikit-learn (Isolation Forest), pandas, joblib
- Visualization: Streamlit, Plotly Express
- Configuration & Data: YAML (config.yaml), JSONL (mock message queues)

5. Engineering Review & Reflection

What worked well:

Decoupling the modules using intermediate queues allowed each component to be developed and tested independently.

What would you do differently:

For a true production environment, I would replace the file-based queues with Apache Kafka or RabbitMQ, and utilize Docker Compose.

Performance & Scalability:

The platform processes events sub-millisecond per log. To scale 10x, deploying Kubernetes pods for horizontal scaling would be required.

Security Vulnerabilities:

Current vulnerabilities include lack of dashboard authentication, on-disk plaintext JSONL queues susceptible to tampering, and the need for a secure secrets manager.

6. Demo Screenshots

Figure: p1.PNG

Real-time Threat Detection and Orchestration Dashboard

1183

850

331

1187

854

331

1191

857

332

1194

860

332

1197

863

332

Go to Settings to activate Windows.

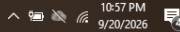


Figure: p2.PNG

Total Events

High Cont

Medium C

1197

863

332

1202

867

333

1206

871

333

1209

873

334

1212

875

335

Activate Windows
Go to Settings to activate Windows.



Figure: p3.PNG

Live Alert Feed

	timestamp	source_ip	destination_domain	ml_prediction	ti_risk_score	combined_risk_score	Risk L
1211	2026-09-20 22:56:44	8.8.8.8	malicious-site.net	-1	0.95	0.98	High
1210	2026-09-20 22:56:43	45.33.32.156	google.com	-1	0.88	0.952	High
1209	2026-09-20 22:56:42	192.168.1.10	example.com	-1	0.05	0.62	Medi
1208	2026-09-20 22:56:42	45.33.32.156	google.com	-1	0.88	0.952	High
1207	2026-09-20 22:56:41	192.168.1.10	suspicious.org	-1	0.95	0.98	High
1206	2026-09-20 22:56:41	8.8.8.8	google.com	-1	0.05	0.62	Medi
1205	2026-09-20 22:56:40	192.168.1.10	suspicious.org	-1	0.95	0.98	High
1204	2026-09-20 22:56:39	8.8.8.8	suspicious.org	1	0.95	0.98	High
1203	2026-09-20 22:56:38	185.15.59.224	example.com	1	0.95	0.9433	High
1202	2026-09-20 22:56:38	45.33.32.156	suspicious.org	-1	0.95	0.98	High

Threat Intelligence

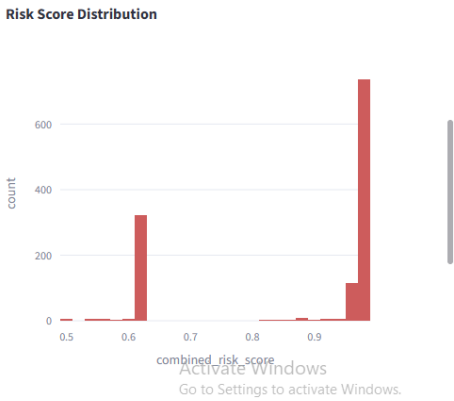


Figure: p4.PNG

Anomaly vs Normal

ML Detections

